

PAGE 58–59: THE THIRD DIMENSION

	FACES	EDGES	CORNERS
CUBE	6	12	8
TETRAHEDRON	4	6	4
OCTAHEDRON	8	12	6
DODECAHEDRON	12	30	20
ICOSAHEDRON	20	30	12

Number of faces + number of corners = number of edges + 2

PAGE 60–61: FOOTBALLS AND BUCKYBALLS

Cube puzzle

You need six cuts, since the central cube has six faces.

PAGE 62–63: ROUND AND ROUND

Rolling coins

Most people think the coin will make a half turn, but in fact it makes a complete turn.

The bear hunter

White. The hunter must be at the North Pole, so it's a polar bear.

Flying tonight

The pilot must have been flying to exactly the other side of the world from the airport. Wherever the girl was going, he could fly past her destination.

The area of the pink band

They're the same area. The radius of the circles increases by 1 unit each time. The area of the three middle rings, therefore, is $\pi 3^2$, or 9π . The area of the blue ring = $\pi 5^2 - \pi 4^2$, which is also 9π .

PAGE 66–67: SHAPES THAT STRETCH

Topological shapes

The doughnut is equivalent to the needle, cotton spool, cup, and

funnel. The rugby ball is equivalent to the glass, football, battery, die, and pencil. The spanner is equivalent to the scissors and bowl. The brick is the odd one out.

The Möbius strip

When you cut along the centre, the Möbius strip turns into one band twice as long. When you cut a third of the way from the edge, the strip turns into two rings linked together.

The two ribbons

A square

Monkey puzzle

Yes. Imagine two people walking up and down the mountain the same day. Whatever their speed, they must meet each other at some point.

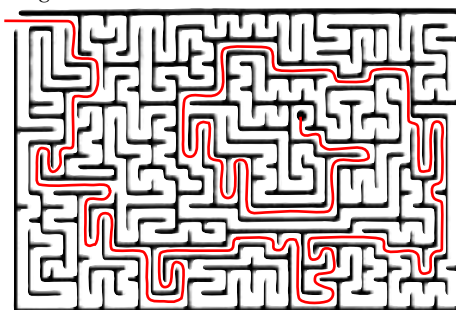
PAGE 68–69: MIRROR MIRROR

Axes of symmetry

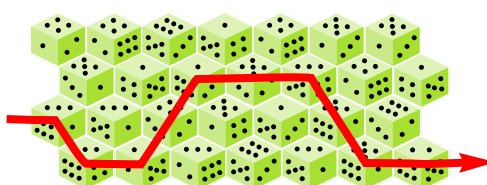
Sellotape: infinite (or 4 if you think the plastic struts matter). Flower: about as many as the number of petals. Star: 5. Bat: 1. Scissors: 1. Crab: 0. Coin: 7. Spoon: 1.

PAGE 70–71: AMAZING MAZES

Big maze



Dice maze



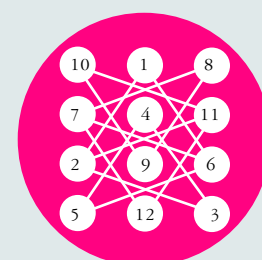
Spiral maze

All the dots are outside the maze. To find out whether a dot is inside or outside, count the number of lines between the dot and outer edge of the maze. An even number means the dot is outside; an odd number means the dot is inside.

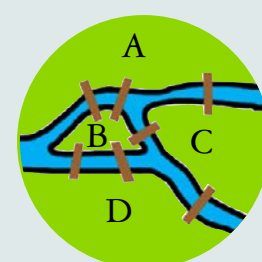
Coloured sticks

Blue, red, pink, pale blue, yellow, pale green, grey, dark green.

Twelve connected circles



Königsberg bridges



The Swiss mathematician Leonhard Euler used network theory to solve this famous puzzle in 1736. Imagine the town as having four regions: A, B, C, and D. On your walk through town, you'd walk in and out of at least two of these regions the same number of times, so they'd have to have an even number of bridges. But all the regions have an odd number of bridges, so the walk must be impossible.